

Agenda

- ① String Introduction
- ② null character
- ③ User input
- ④ string functions
- ⑤ Function call by passing string
- ⑥ Handling multiple strings.

String

- String is a sequence of characters, terminated at null character.
- Strings are handled in char arrays.

```
char str[10];
```

Initializing char array during declaration

```
int main()  
{
```

```
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };  
    int i;
```

```
    for(i=0; i<=9; i++)
```

```
        printf("%c", str[i]);
```

```
    return 0;
```

```
}
```

0 1 2 3 4 5 6 7 8 9

B H O P A L \0 \0 \0 \0

' '\0 0

' ' 32

'A' 65 A

'B' 66 B

Printing String

```
int main()  
{
```

```
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
```

```
    int i;
```

```
    for(i=0; str[i] ; i++)
```

```
        printf("%c", str[i]);
```

```
    return 0;
```

```
}
```

~~$i \neq '\backslash 0'$
 $i < '\backslash 0'$
 $i \leq \text{str}[i]$
 $i \neq \text{NULL}$~~

$i = 0, 1, 2, 3, 4, \dots$
 $\text{str}[i] = 'B', 'H', 'O', 'P', \dots$
 $\dots '\backslash 0'$

Improve your code

```
int main()
```

```
{
```

```
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
```

```
    int i;
```

```
    for (i=0; str[i]; i++)
```

```
        printf("%c", str[i]);
```

```
}
```

Use of null character

```
int main()  
{
```

```
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
```

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

B	H	O	P	A	L	\0			
---	---	---	---	---	---	----	--	--	--

```
}
```

%s

```
int main()
```

```
{
```

```
char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
```

```
printf("%s", str);
```

0	1	2	3	4	5
B	H	O	P	A	L
C	I	P	Q	B	M

```
}
```

i	str[i]
0	B
1	H
2	O

str[i] = str[i] + 1
 'B' + 1

t = 1

Calculating length of the String

0 1 2 3 4 5 6
B H O P A L \0

```
int main()
```

```
{
```

```
    char str[10] = "BHOPAL";
```

```
    int i;
```

```
    for (i=0; str[i]; i++);
```

```
    printf("Length is %d", i);
```

```
}
```

String Constant

"BHOPAL" ← String constant
string Literal

char str[10] = "BHOPAL";

0	1	2	3	4	5	6	7	8	9
B	H	O	P	A	L	\0			

Taking input from User

- scanf is not capable to input `scanf()` multiword string
- because space, tab, new line characters are delimiters
- we will not use scanf for string input

`gets()`

- `gets()` is capable to input multiword string

gets()

only for strings
one string at a time

scanf()

multiple values

MySirG Education

Services Private Limited

↑
unreliable

fgets()

`fgets( arrayname , inputsize , stdin )`

Memory Concept

(byte = 8 bits)

```
int i;
```

```
char str[20] = "My Sir G";
```

option 1

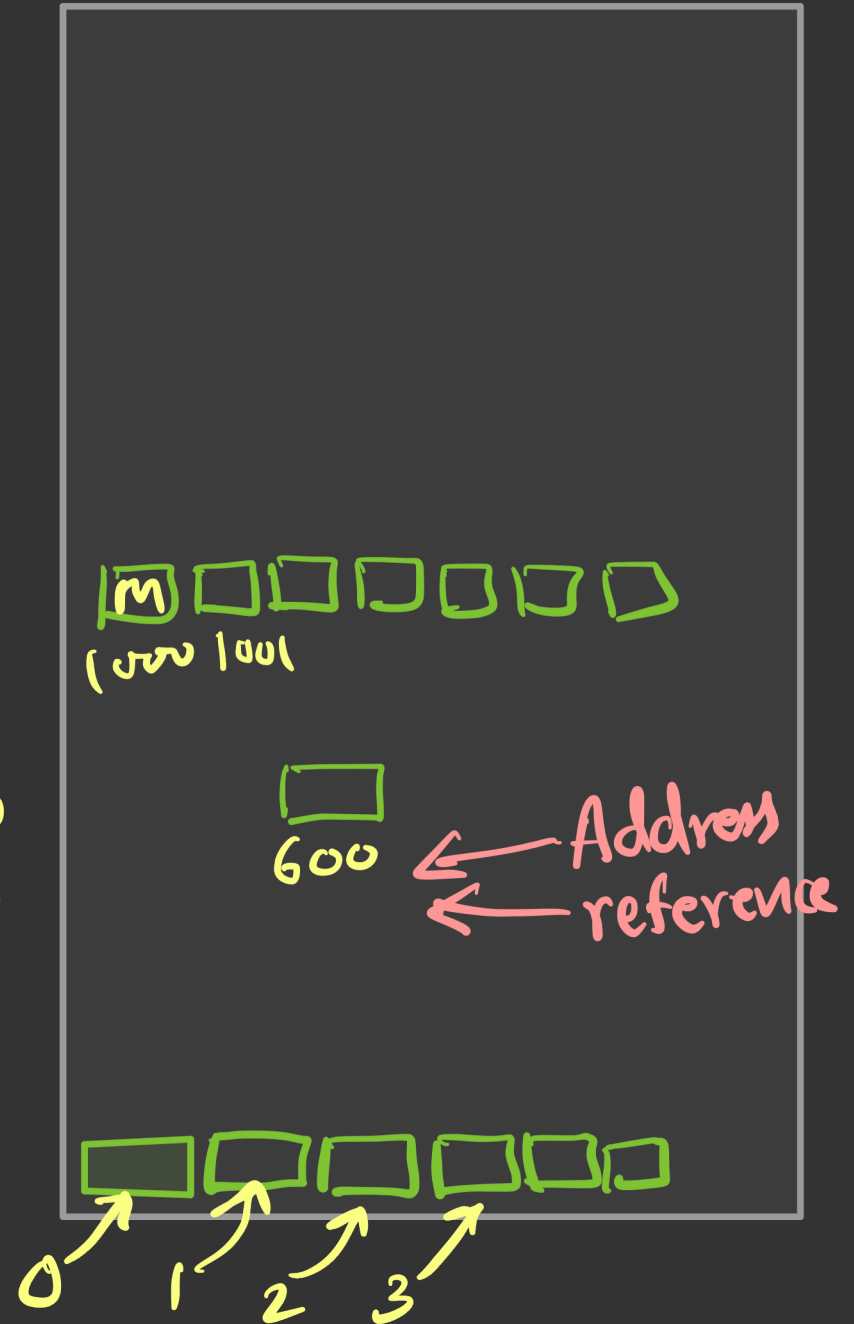
```
for (i=0; str[i]; i++)  
    printf("%c", str[i]);
```

option 2

```
printf("%s", str);
```

```
&str[0] ≈ 1000  
&str[1] ≈ 1001  
char x;
```

```
str[0] 'M'  
str == &str[0] 1000
```



String Functions

```
#include <string.h>
```

```
int strlen(char*);
```

```
char* strcpy(char*);
```

```
char* strlwr(char*);
```

```
char* strrev(char*);
```

```
char* strcpy(char*, char*);
```

```
char* strcat(char*, char*);
```

```
int strcmp(char*, char*);
```

```
char s1[] = "Hello"
```

```
char s2[] = "Students"
```

```
l = strlen(str);
```

```
l = strlen("Hello");
```

```
printf("%s", strcpy(str));
```

```
strlwr(str);
```

```
strrev(str);
```

```
strcpy(str, "BHOPAL");
```

```
strcat(s1, s2);
```

s1 → "Hello Students"

$r = \text{strcmp}(\text{"Amit"}, \text{"Amar"});$

$r = -1$ dictionary order

$r = 0$ equal

$r = 1$ opp of dictionary order

Function call by passing String

```
f1("Bhopal");  
f1(st);
```

```
void f1(char s[])  
{  
  
}
```

& s[0][0] == Handling Multiple Strings

char s[4][10] = { "Pune", "Bhopal", "Bengaluru",
"Patna" };

	0	1	2	3	4	5	6	7	8	9
s[0]	0	P	u	n	e					
s[1]	1	B	h	o	p	a	l			
s[2]	2	B	e	n	g	a	l	u	r	
s[3]	3	P	a	t	n	a				

```
for (i=0; i<=3; i++)  
    printf("%s", s[i]);
```

```
char s[5][20];
```

```
int i;
```

```
printf("Enter 5 city names");
```

```
for(i=0; i<=4; i++)
```

```
    fgets(s[i], 20, stdin);
```

Write a recursive function to print binary of a given decimal number.

```
void dtob(int x)
```

```
{
```

```
    if (x > 0)
```

```
    {
```

```
        dtob(x/2);
```

```
        printf("%d", x%2);
```

```
    }
```

```
}
```

RC ① dtob(25) 11001

RC ② dtob(25/2) 1100
printf("%d", 25%2); 1

BC ③ $x > 0$

13

①

②

$$n + \text{sum}(n-1)$$

$$2 * n - 1 + \text{sum}(n-1)$$

1 2 3 4 ... n

1 3 5 7 ... $2n-1$

③

$$2 * n + \text{sum}(n-1)$$

$$n * n + \text{sum}(n-1)$$

$$\text{sum}(n/10) + n \% 10$$

④

⑤

⑥

$$n * \text{fact}(n-1)$$

2451

$$2 + 4 + 5 + 1$$

$$\textcircled{1} \text{sum}(2451) \quad 12$$

$$\textcircled{2} \text{sum}(245) \quad 11 \\ + 2451 \% 10$$

⑦ HCF

$$(245, 125)$$

$$(245 \% 125, 125)$$

$$(120, 125)$$

$$(120, 125 \% 120)$$

$$(120, 5)$$

a b

$\text{if } (a \% b == 0)$
 $\text{return } b;$